

**VIETNAM GENERAL CONFEDERATION OF LABOR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



**VO XUAN QUANG – 523H0173
HOANG XUAN THANH – 523H0178**

FINAL REPORT

COMPUTER VISION

HO CHI MINH CITY, 2026

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ViGround-Contrast: Low-Resource Vietnamese Referring Expression Grounding

Advised by

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HO CHI MINH CITY, 2026

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We would like to express our sincere gratitude to Ph.D. Pham Van Huy, our instructor and mentor, for his valuable guidance and support throughout the final report of our project. He has been very helpful and patient in providing us with constructive feedback and suggestions to improve our work. He has also encouraged us to explore new technologies and techniques to enhance our system's functionality and performance. We have learned a lot from his expertise and experience in computer vision, visual grounding, experimental design, and model evaluation. We are honored and privileged to have him as our teacher and supervisor.

Ho Chi Minh city, 18th Jun 2026.

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DECLARATION OF AUTHORSHIP

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ABSTRACT

This study investigates Vietnamese referring-expression grounding through ViGround-Contrast, a bilingual benchmark and controlled low-resource adaptation framework built from RefCOCO. We construct aligned English–Vietnamese standard examples and same-category hard pairs designed to expose instance-confusion errors. Using LocateAnything-3B as the base model, we compare two LoRA adaptation strategies under an identical budget of 4,000 Vietnamese training expressions: distribution-matched random sampling and hard-pair-based sampling. We additionally evaluate Grounding DINO and Qwen2.5-VL-3B as cross-architecture zero-shot baselines.

LocateAnything-3B exhibits a 16.6-point English–Vietnamese Acc@0.5 gap on the standard benchmark. Random-FT and HardPair-FT achieve small descriptive Vietnamese gains of 1.5 and 1.6 points, respectively, although their confidence intervals include zero. On the Vietnamese hard-pair benchmark, both fine-tuned models increase pair accuracy from 50.4% to 56.4% and reduce same-box-collapse errors. However, HardPair-FT does not significantly outperform Random-FT, indicating that hard-pair data selection alone is insufficient to exploit pair structure under the current training objective. Cross-architecture evaluation shows that LocateAnything-3B significantly outperforms Qwen2.5-VL-3B and Grounding DINO on Vietnamese grounding, despite Grounding DINO achieving comparable English standard performance. These findings reveal a substantial architecture-dependent Vietnamese grounding gap and motivate future pair-aware contrastive objectives.

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CHAPTER 1. INTRODUCTION

1.1 Background and Problem Statement

Visual grounding requires a model to map a natural-language query to the spatial extent of the referenced object within an image. Because the task couples language understanding with localized visual discrimination, performance often degrades when the query language shifts away from the distribution emphasized during pretraining or earlier evaluation. In multilingual settings, this creates a practical asymmetry: systems that perform strongly on English queries may still be brittle on lower-resource or less represented languages such as Vietnamese.

1.2 The ViGround-Contrast Framework and Research Objectives

Our study introduces the ViGround-Contrast framework to address a specific research question rather than just general adaptation. Our comparison is designed to test whether Vietnamese fine-tuning should simply increase language alignment, or whether our harder sampling regime can additionally reduce the class of errors that arise when multiple visually similar instances compete for the same description.

1.3 Scope and Report Structure

Accordingly, this report adopts a paper-style analytical structure. We detail our experimental design, state what can and cannot be inferred from our evaluation evidence, and argue for conclusions that are tightly bounded by our metrics. The emphasis throughout is on causal restraint: improvements are attributed only where our benchmarks support them, and stronger claims are withheld where confidence intervals or paired comparisons do not justify them.

1.4 Our Difference from Existing Approaches

To the best of our knowledge, ViGround-Contrast distinguishes itself from existing literature in three primary ways. First, to the best of our knowledge, ViGround-Contrast is among the first Vietnamese visual grounding benchmarks specifically designed to study same-category instance confusion. While previous datasets focus on general referring expression comprehension, ours explicitly constructs hard pairs to evaluate instance-level ambiguity and same-box-collapse errors.

Second, it employs a strictly controlled single-variable ablation setup. Rather than arbitrarily sampling a control group, our random control split is carefully matched to the hard pairs by category, sentence length, and bounding box area, ensuring that any observed differences are isolated entirely to the data sampling strategy.

Third, our study moves beyond standard point-estimate evaluations by providing rigorous statistical validation. We report bootstrap confidence intervals and statistical tests for the primary paired comparisons, ensuring that all claims regarding cross-lingual adaptation and instance discrimination are grounded in statistical significance.

CHAPTER 2. RELATED WORK

2.1 Visual Grounding with Deep Architectures

Visual grounding has seen substantial advancements through the development of deep learning architectures capable of multimodal alignment. Early approaches modularized the attention mechanism to parse referring expressions and localize target regions, as seen in MAttNet [5]. More recent frameworks have shifted towards transformer-based architectures, such as TransVG [1], which demonstrated the effectiveness of end-to-end multimodal understanding using transformer encoders.

2.2 Multilingual and Cross-Lingual Grounding

To address language barriers, the field has increasingly explored multilingual grounding solutions like mDETR [2], which provides multilingual modulated detection to generalize referring expression comprehension. Concurrently, large vision-language foundation models have set new state-of-the-art standards. Grounding DINO [8] has successfully married vision transformers with grounded pre-training for robust open-set object detection. Meanwhile, highly capable multimodal LLMs such as Qwen2.5-VL and LocateAnything-3B [9] have demonstrated remarkable zero-shot reasoning and spatial localization capabilities across varied tasks.

2.3 Parameter-Efficient Adaptation and Contrastive Sampling

Despite these advancements, existing literature leaves critical gaps that ViGround-Contrast directly addresses. First, while foundation models excel in general tasks, there is an absence of bilingual benchmarks (EN-VI) tailored specifically to expose instance-level ambiguity for lower-resource languages. Second, current models often lack a data-driven focus on resolving instance-confusion (hard pairs) when multiple visually identical objects are present. Finally,

our study differs by providing a strictly controlled, single-variable ablation setup using an identical budget of 4,000 samples, moving beyond general benchmarking to isolate the exact impact of hard-pair data sampling.

2.4 Limitations of Prior Work and Points of Differentiation

The existing literature leaves several critical gaps. First, there is an absence of bilingual benchmarks tailored to expose instance-level ambiguity for lower-resource languages like Vietnamese. Second, adaptation studies often lack tightly controlled ablations that isolate data sampling strategies from model architecture or scale. Finally, there is an over-reliance on point-estimate evaluations without rigorous statistical significance testing.

ViGround-Contrast directly maps its contributions to these limitations. We provide the necessary bilingual hard-pair benchmark to test instance confusion, implement a single-variable experimental design that matches random controls to hard pairs, and validate all observed performance shifts using strict bootstrap statistical testing.

CHAPTER 3. DATASET CONSTRUCTION AND EXPERIMENTAL PIPELINE

3.1 Training Setup

We utilized the LocateAnything-3B architecture to train two distinct models under identical pipelines. The primary difference between our `viground-random-ft-locany` and `viground-hardpair-ft-locany` branches lies strictly in the construction and sampling of the training data. The consistent directory structures and hyperparameter settings ensure that our comparison reflects the impact of data sampling rather than optimization artifacts.

3.2 Evaluation Artifacts and Chain of Evidence

The key comparative evidence resides in the dataset artifact and benchmarks we constructed (**viground-contrast-results**). This artifact contains our benchmark tables, hard-pair metrics, paired-effects files, latency summaries, training configuration files, and failure-case outputs. This separation between checkpoints and evaluation artifacts is beneficial for analysis because it makes the chain of evidence transparent: the model checkpoints indicate what we trained, whereas the results artifact indicates how we measured performance.

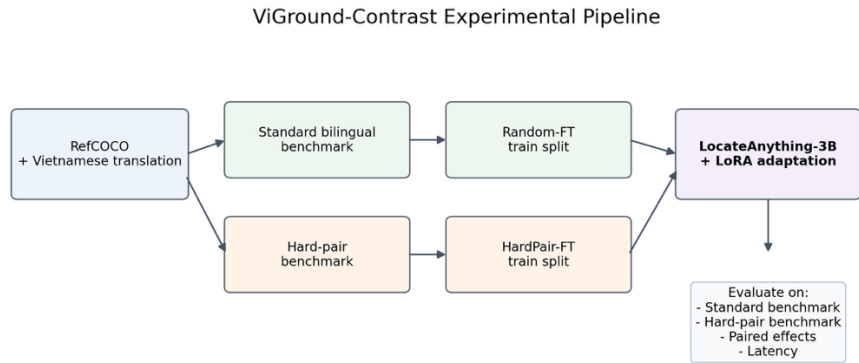


Table 1: Overview of the ViGround-Contrast experimental pipeline

CHAPTER 4. PROBLEM STATEMENT AND RESEARCH OBJECTIVE

4.1 Problem Formulation and Evaluation Challenges

The task considered here is visual grounding with image-language input pairs. Given an image and a query, the system predicts a bounding box intended to match the ground-truth referent. In ordinary benchmark settings, the prediction is evaluated by overlap-based metrics such as mIoU and thresholded accuracy. In hard-pair settings, however, correct performance requires not only locating a plausible object of the right category, but selecting the correct instance among multiple competing alternatives.

4.2 Nested Research Objectives

The research objective can therefore be stated in three nested layers. First, determine whether Vietnamese fine-tuning improves Vietnamese grounding relative to the base LocateAnything-3B model. Second, determine whether hard-pair fine-tuning yields any advantage over random fine-tuning when the benchmark is designed to expose instance-level ambiguity. Third, characterize the trade-off between Vietnamese specialization, English retention, and inference cost. A rigorous answer requires examining not only aggregate success metrics but also the error categories that motivated hard-pair construction in the first place.

CHAPTER 5. DATA AND BENCHMARK DESIGN

5.1 Dataset Splits and Experimental Statistics

Our dataset provides enough information to reconstruct the experimental data regime at a high level. The training and evaluation splits are summarized in **Table 2**.

Split	Samples	Pairs	Unique images	Categories	Language	Translation valid rate
train random	4000	-	3083	69	Vi	0.99775
train_hardpair	4000	2000	1513	69	Vi	0.99875
standard_benchmark	2000	-	826	61	En, Vi	0.99
hard_pair_benchmark	1000	500	356	60	En, Vi	-

Table 2 : Released training and evaluation split statistics from the viground – contrast - results artifact.

5.2 Methodological Implications

Several implications follow directly from these counts. First, the comparison between Random-FT and HardPair-FT is not confounded by raw sample count, because both training sets contain 4000 samples. Second, the hard-pair training set uses far fewer unique images while encoding 2000 explicit pairs, which implies that its signal is concentrated on ambiguous within-image discrimination rather than broader image diversity. Third, the evaluation protocol separates ordinary benchmark behavior from adversarial structured pair behavior, allowing the study to ask whether a model that improves on general Vietnamese grounding also improves on the more specific phenomenon of instance confusion.

To build our framework, we utilized data from RefCOCO UNC and images from MS COCO 2014, joining them effectively by `image_id`. Our process involved rigorous bounding box conversions, English-Vietnamese alignment, and automated translation with validation and repair processes. Crucially, we ensured that the train-benchmark leakage was strictly zero. The hard-pair training set uses far fewer unique images while encoding 2,000 explicit pairs, which implies that its signal is concentrated on ambiguous within-image discrimination rather than broader image diversity.

CHAPTER 6. METHOD

6.1 Base Model and Controlled Comparison

Both released fine-tuned models are built on the same base model, NVIDIA/LocateAnything-3B. This design choice is crucial because it narrows the space of plausible explanations for any observed difference. Since the architecture and base weights are shared, the most defensible attribution is that performance differences arise from the fine-tuning data regime rather than from unrelated changes in model family or inference behavior.

6.2 Random Fine-Tuning versus Hard-Pair Fine-Tuning

Random-FT represents a conventional adaptation baseline: Vietnamese training examples are sampled without explicitly privileging referential ambiguity. HardPair-FT instead concentrates learning on cases where different object instances are easy to confuse. In principle, this should not necessarily maximize average benchmark accuracy on all examples; rather, it should alter the model's discrimination behavior in those cases where ordinary language alignment is insufficient. The correct theoretical expectation is therefore subtle: if hard-pair training works, it may change the pattern of failures before it changes the top-line aggregate metric.

6.2.1 *Hard-Pair Mining:*

We extracted challenging samples by requiring the same `image_id` and category, but explicitly different `ann_id` and bounding boxes. The text queries had to share similar expressions but differ in distinguishing attributes. We also maintained sample uniqueness and set a strict limit on the number of pairs per image

6.2.2 Random Control Matching

Our Random-FT baseline was not arbitrarily sampled. It was strictly matched to the hard-pair distribution by category, sentence length, bounding box area, and spatial rate to ensure a fair and controlled comparison.

6.3 Training Configuration

The released resolved configuration files show that both branches were trained under nearly identical settings:

- Epochs: 1
- Optimization steps: 500
- Per-device batch size: 1
- Gradient accumulation: 8
- Learning rate: $2e-6$ (cosine schedule)
- Weight decay: 0.01
- Precision: bf16
- Maximum sequence length: 2048

The LoRA [17]-related configuration indicates lightweight adaptation rather than full dense retraining. This matters for interpretation because it means the released comparison is not between radically different optimization budgets; it is between two sampling strategies under a fixed adaptation budget.

6.4 Released Training Outcomes

The model repositories additionally expose training summaries. Random-FT reports a final training loss of 1.2232 and a runtime of 20090.58 seconds, while HardPair-FT reports a lower final training loss of 1.1359 with a similar runtime of 20057.11 seconds. These values show that HardPair-FT fit its training data more easily or more effectively under the same budget. Yet lower training loss alone is not

evidence of superior task performance; it only establishes that the optimization trace was favorable on the training objective. Whether that advantage transfers to the released evaluation tasks must be judged separately.

CHAPTER 7. EVALUATION PROTOCOL

7.1 Standard Benchmark Metrics

The evaluation protocol has two complementary layers. The standard benchmark reports **mIoU**, **Acc@0.5**, **Acc@0.75**, **parse failure**, and **multi-box failure**. These metrics capture the quality and stability of predicted boxes under conventional grounding evaluation. They are suitable for testing whether Vietnamese fine-tuning improves general grounding alignment.

7.2 Hard-Pair Benchmark Metrics

The hard-pair benchmark adds pair-level metrics, including pair accuracy, pair mIoU, wrong-instance rate, and same-box-collapse rate. These metrics are better aligned with the paper's deeper hypothesis. If hard-pair training is effective, it should reduce the tendency to assign both queries to the same box or to choose the wrong instance among otherwise plausible candidates. Therefore, the hard-pair benchmark is not merely an auxiliary test set; it is the primary instrument for evaluating whether the hard-pair intervention works for the reason it was introduced.

7.3 Statistical Significance and Paired Effects

The artifact also includes paired-effects statistics with bootstrap confidence intervals and p-values for selected comparisons. This is important because small benchmark gains can be overinterpreted if only point estimates are reported. The statistical files make it possible to distinguish between descriptive improvements and differences that are strongly supported as stable under resampling.

7.4 Cross-Architecture Evaluation Setup

For our cross-architecture evaluation, both Grounding DINO and Qwen2.5-VL-3B were evaluated in a strictly zero-shot setting to establish baseline performance. Grounding DINO outputs were processed by selecting the highest-confidence proposal using a fixed threshold. Qwen2.5-VL-3B predictions, originally generated as pixel coordinates, were subsequently converted into our standardized bounding box coordinate system. To ensure absolute fairness, both models were evaluated using the exact same images, queries, ground-truth bounding boxes, and IoU evaluator pipeline as our base and fine-tuned models.

CHAPTER 8. EXPERIMENTAL RESULTS

8.1 Standard Benchmark Results

Model	Lang	mIoU	Acc@0.5	Acc@0.75	Parse fail	Multi-box	Grounding gap
base	En	0.8027	0.870	0.792	0.080	0.080	-
base	Vi	0.6627	0.704	0.627	0.059	0.056	0.166
random_ft	En	0.7779	0.842	0.770	0.108	0.108	-
random_ft	Vi	0.6758	0.719	0.643	0.067	0.060	0.123
hardpair_ft	En	0.7715	0.834	0.762	0.114	0.114	-
hardpair_ft	Vi	0.6774	0.720	0.648	0.069	0.062	0.114

Table 3 : Released standard benchmark results for the base model and the two fine - tuned models.

The standard benchmark results show that fine-tuning produced small descriptive improvements on Vietnamese standard grounding, but the gains were not statistically significant under the 4,000-sample training budget. On Vietnamese queries, Acc@0.5 increases from 0.704 for the base model to 0.719 for Random-FT and 0.720 for HardPair-FT. Likewise, mIoU increases from 0.6627 to 0.6758 and 0.6774, respectively. The English-Vietnamese grounding gap narrows substantially, from 0.166 in the base model to 0.123 in Random-FT and 0.114 in HardPair-FT.

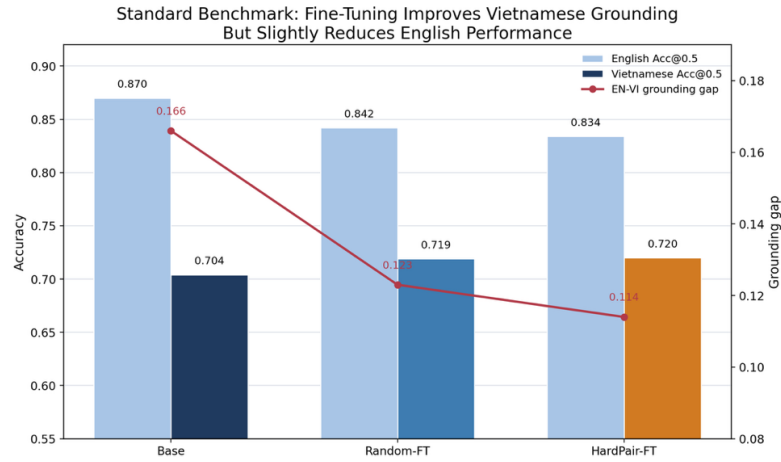


Figure: Standard benchmark results

At the same time, the same table supports a second and more cautionary claim: Vietnamese adaptation is not free. English **Acc@0.5** falls from 0.870 in the base model to 0.842 for Random-FT and 0.834 for HardPair-FT, while parse-failure and multi-box-failure rates rise. Therefore, the released evidence does not support the interpretation that fine-tuning simply makes the model better in a language-agnostic sense. A stricter reading is that it reallocates model behavior toward the Vietnamese target distribution, reducing cross-lingual imbalance while sacrificing some English performance.

8.2 Hard-Pair Benchmark Results

Model	Lang	Pair acc	Pair mIoU	Wrong-instance	Same-box collapse	Parse fail
base	Vi	0.504	0.4869	0.119	0.124	0.031
random_ft	Vi	0.564	0.5317	0.101	0.086	0.036
hardpair_ft	Vi	0.564	0.5317	0.095	0.082	0.035
base	En	0.798	0.7137	0.028	0.020	0.047
random_ft	En	0.782	0.6945	0.027	0.026	0.062
hardpair_ft	En	0.786	0.7007	0.026	0.024	0.057

Table 4: Released hard-pair benchmark results, which are most directly aligned with the hard - pair training hypothesis.

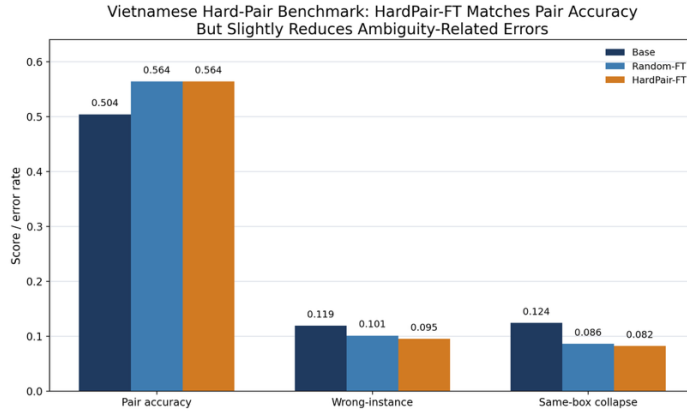


Figure: Vietnamese hard-pair benchmark results

The hard-pair benchmark yields the core substantive result of the project. On Vietnamese pairs, both fine-tuned models increase pair accuracy from 0.504 to 0.564, a nontrivial gain over the base model. However, HardPair-FT does not exceed Random-FT on this top-line metric; both are tied at 0.564. If the analysis stopped here, the appropriate conclusion would be that hard-pair sampling offers no measurable benefit over ordinary Vietnamese fine-tuning on the main pair-level accuracy statistic.

Yet that conclusion would be incomplete, because it would ignore the exact failure mechanisms that the hard-pair intervention was designed to change. Relative to Random-FT, HardPair-FT lowers the Vietnamese wrong-instance rate from 0.101 to 0.095 and lowers same-box collapse from 0.086 to 0.082, while leaving pair accuracy and pair mIoU unchanged. This means the released evidence is consistent with a narrower interpretation: hard-pair fine-tuning may improve how the model fails, even when it does not improve how often it succeeds under the chosen aggregate metric. In other words, the intervention appears to reshape the error distribution more clearly than it reshapes the headline score.

The English hard-pair results reinforce the adaptation trade-off observed earlier. Both fine-tuned models are slightly worse than the base model on English pair accuracy, and neither gains a decisive advantage there. Thus, the artifact supports specialization rather than universal improvement.

Model	VI Standard Acc@0.5	VI Pair Accuracy
HardPair-FT	0.720	0.564
Random-FT	0.719	0.564
LocateAnything Base	0.704	0.504
Qwen2.5-VL-3B	0.619	0.408
Grounding DINO	0.342	0.030

Table 5: Cross-Architecture Evaluation on Vietnamese Benchmarks.

8.3 Statistical Evidence and Inferential Limits

Comparison	Metric	N	Abs. diff.	95% CI	p-value
Random-FT vs Base	Acc@0.5	1000	+0.015	[-0.002, 0.032]	0.11466
HardPair-FT vs Base	Acc@0.5	1000	+0.016	[-0.001, 0.033]	0.080507
HardPair-FT vs Random-FT	Pair accuracy	500	0.000	[-0.016, 0.016]	1.0
HardPair-FT vs Random-FT	Wrong-instance	500	-0.006	[-0.016, 0.004]	n/a
HardPair-FT vs Random-FT	Same-box collapse	500	-0.004	[-0.020, 0.010]	0.790527

Table 6 : Released paired-effects summary with bootstrap confidence intervals and p-values for selected comparisons.

Comparison	Metric	Difference	95% CI	p-value
DINO vs Base	VI Acc@0.5	-0.362	[-0.401, -0.324]	<0.0001
DINO vs Base	VI Pair Accuracy	-0.474	[-0.522, -0.428]	<0.0001
Qwen vs Base	VI Acc@0.5	-0.085	[-0.124, -0.047]	0.000017
Qwen vs Base	VI Pair Accuracy	-0.096	[-0.150, -0.042]	0.000655

Table 7: Statistical Evaluation for Cross-Architecture Baselines.

These statistics materially constrain what may be claimed. The gains of Random-FT and HardPair-FT over the base model on Vietnamese Acc@0.5 are directionally positive, but the confidence intervals cross zero and the p-values do not provide strong conventional evidence of a stable nonzero gain under the released resampling procedure. This does not invalidate the descriptive improvements, but it does mean they should be phrased as suggestive rather than definitive.

More importantly, the direct comparison that matters most for the paper's central claim HardPair-FT versus Random-FT on Vietnamese pair accuracy shows an absolute difference of exactly 0.000 with a 95% confidence interval of

$[-0.016, 0.016]$ and a p-value of 1.0. That result sharply limits any attempt to argue that hard-pair training improves the main hard-pair score. The most defensible formulation is therefore narrower: HardPair-FT does not improve the released aggregate pair-accuracy metric relative to Random-FT, but it does move two targeted error metrics in the expected direction.

8.4 Inference Cost

Latency measurements add another practical dimension. The base model reports a mean latency of approximately **1459 ms** and a median latency of approximately **467 ms** across the full evaluation set. Both fine-tuned models increase mean latency to roughly **2064-2065 ms** and median latency to roughly **544 ms**. Since Random-FT and HardPair-FT are almost indistinguishable in latency, inference cost does not help choose between them; it only reinforces that adaptation incurs nontrivial runtime overhead relative to the base model.

CHAPTER 9. DISCUSSION

9.1 Validation of Vietnamese Fine-Tuning and Grounding Gap

Both fine-tuned models achieve descriptive Vietnamese gains of 1.5 points (Random-FT) and 1.6 points (HardPair-FT) on $\text{Acc}@0.5$ relative to the base model. However, the 95% bootstrap confidence intervals for both gains include zero ($p = 0.115$ and $p = 0.081$, respectively), indicating that these improvements are not statistically significant under our 4,000-sample training budget. Consequently, the correct reading is that fine-tuning moves Vietnamese performance in the right direction without yet establishing a reliable, replicable gain. The English–Vietnamese grounding gap narrows from 16.6 points at baseline to 12.3 points (Random-FT) and 11.4 points (HardPair-FT). Importantly, this gap reduction is only partly attributable to Vietnamese improvement; it is also driven by a concurrent drop in English $\text{Acc}@0.5$ from 0.870 to 0.842 and 0.834. The gap therefore closes from both sides, and any interpretation of gap reduction as unambiguous progress must account for this English regression.

9.2 Efficacy of Hard-Pair Sampling

Second, the evidence does not support a stronger claim that hard-pair fine-tuning outperforms random fine-tuning on the released main hard-pair metric. The direct paired comparison on pair accuracy is exactly neutral.

Third, the hard-pair intervention is nevertheless not vacuous. Its most plausible effect is not to raise the overall Vietnamese pair-accuracy ceiling under the released budget, but to alter the kinds of mistakes the model makes. The decreases in wrong-instance and same-box-collapse rates are small, yet they are directionally coherent with the design motivation of hard-pair sampling. A careful interpretation is therefore that the intervention acts more on referential precision than on aggregate success frequency.

9.3 Multilingual Specialization Trade-Off

Fourth, the released setup exhibits a specialization trade-off. The same fine-tuning that helps Vietnamese grounding also weakens English performance. This trade-off should not be treated as an incidental side effect; it is part of the substantive result. The artifacts indicate that, under this budget and data regime, the model reallocates capacity toward the Vietnamese target distribution rather than achieving uniformly better multilingual grounding.

9.4 Explaining the Generalization Gap

This reading also explains why HardPair-FT can show lower training loss without yielding a higher released pair-accuracy score. The model may have learned a more discriminative internal representation for ambiguous training cases, yet the benchmark size, the 500-step training budget, and the metric granularity may be insufficient to convert that representational shift into a larger aggregate advantage. In that sense, the project demonstrates a plausible mechanism without fully proving a broad performance gain.

CHAPTER 10. LIMITATIONS AND THREATS TO VALIDITY

10.1 Data and Training Scale Constraints

This study is subject to several specific constraints regarding data and training scale. First, Vietnamese expressions were primarily generated through automatic translation and have not been fully human-verified across the entire dataset, meaning that annotation noise may partially explain the non-significant $\text{Acc}@0.5$ gains. Second, fine-tuning was strictly constrained to a low-resource budget of only 4,000 expressions per model, which limits the statistical power available to detect modest but real improvements. Third, the optimization process was confined to a single epoch under a single LoRA configuration, leaving open the possibility that extended training or alternative adapter designs could produce stronger results. Finally, the current framework trains on hard pairs as independent examples rather than as structured pairs; it lacks a pair-aware or contrastive loss that could explicitly force the model to separate ambiguous instances within the same training batch. Addressing these constraints represents the most direct path toward stronger and more replicable conclusions.

10.2 Statistical Power and Benchmark Size

A second limitation concerns statistical power and benchmark size. The hard-pair benchmark contains 500 pairs, which is substantial enough to reveal descriptive trends but may be too small to detect modest improvements with high confidence. As a result, the absence of a statistically strong gain for HardPair-FT over Random-FT should not be overread as evidence that the intervention is useless; it is better read as evidence that the released experiment does not yet establish a reliable top-line advantage.

10.3 Model Specificity and Generalization Limits

A third limitation is model specificity. All conclusions are tied to the behavior of LocateAnything-3B under a particular lightweight fine-tuning setup. The released evidence does not justify generalizing the same conclusions to other grounding architectures without additional experiments.

CHAPTER 11. CONCLUSION

11.1 Overall Adaptation Success

Our experiments demonstrate that Random-FT and HardPair-FT provide small descriptive improvements in Vietnamese grounding, though without strong statistical significance under our training budget. Additionally, our cross-architecture evaluation reveals a substantial architecture-dependent Vietnamese grounding gap, as LocateAnything-3B significantly outperformed both Qwen2.5-VL-3B and Grounding DINO.

11.2 The Specific Impact of Hard-Pair Training

While HardPair-FT did not significantly outperform Random-FT on the aggregate pair accuracy, it produced small directional reductions in wrong-instance and same-box-collapse error rates. The project provides evidence for a targeted behavioral effect, indicating that hard-pair sampling alters the error profile in the desired direction, even if it does not yet yield a decisive top-line superiority claim.

CHAPTER 12. FUTURE WORK

12.1 Scaling Benchmarks and Optimization

The most immediate next step is to strengthen the inferential case around hard-pair training. This would require larger hard-pair benchmarks, more ablations on the ratio of random to hard-pair samples, and longer or more varied optimization budgets.

12.2 Qualitative Failure Analysis

It would also be valuable to add a more systematic qualitative analysis of failure cases so that the benchmark-level reductions in wrong-instance and same-box-collapse behavior can be visually tied to concrete prediction improvements.

12.3 Pair-Aware Contrastive Training

Future work should jointly train both members of each hard pair and introduce a pair-aware or contrastive objective that explicitly penalizes wrong-instance selection and same-box collapse. The current framework treats hard-pair examples as independent training samples, which means the model never sees both members of a pair in the same optimization step and therefore cannot learn to explicitly discriminate between them. A contrastive loss that brings together the correct (query, instance) pair while pushing apart the confusable alternative would directly target the failure modes that motivate this work and is the most promising single extension to the present setup.

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